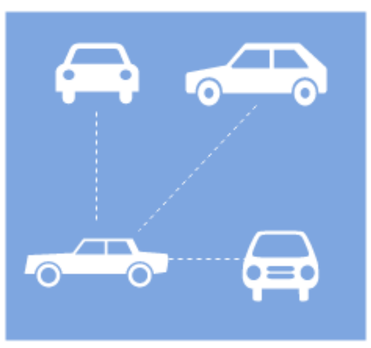
RxSpatial: Reactive Spatial Library for Real-Time Location Tracking and Processing

Youying Shi Abdeltawab Hendawi Hossam Fattah Mohamed Ali

Motivation



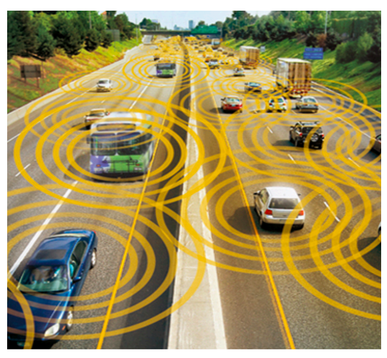
Collaborative Vehicle Cluster



Geofencing

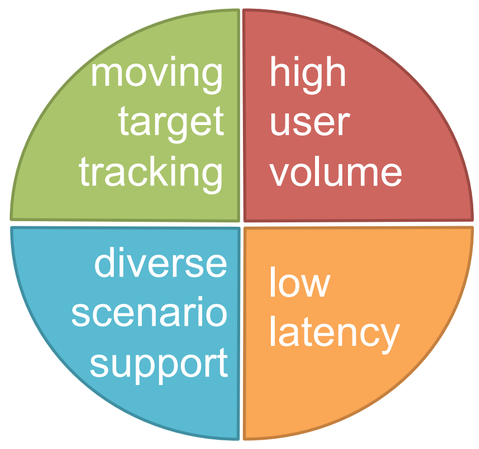


Offender Activity Tracking



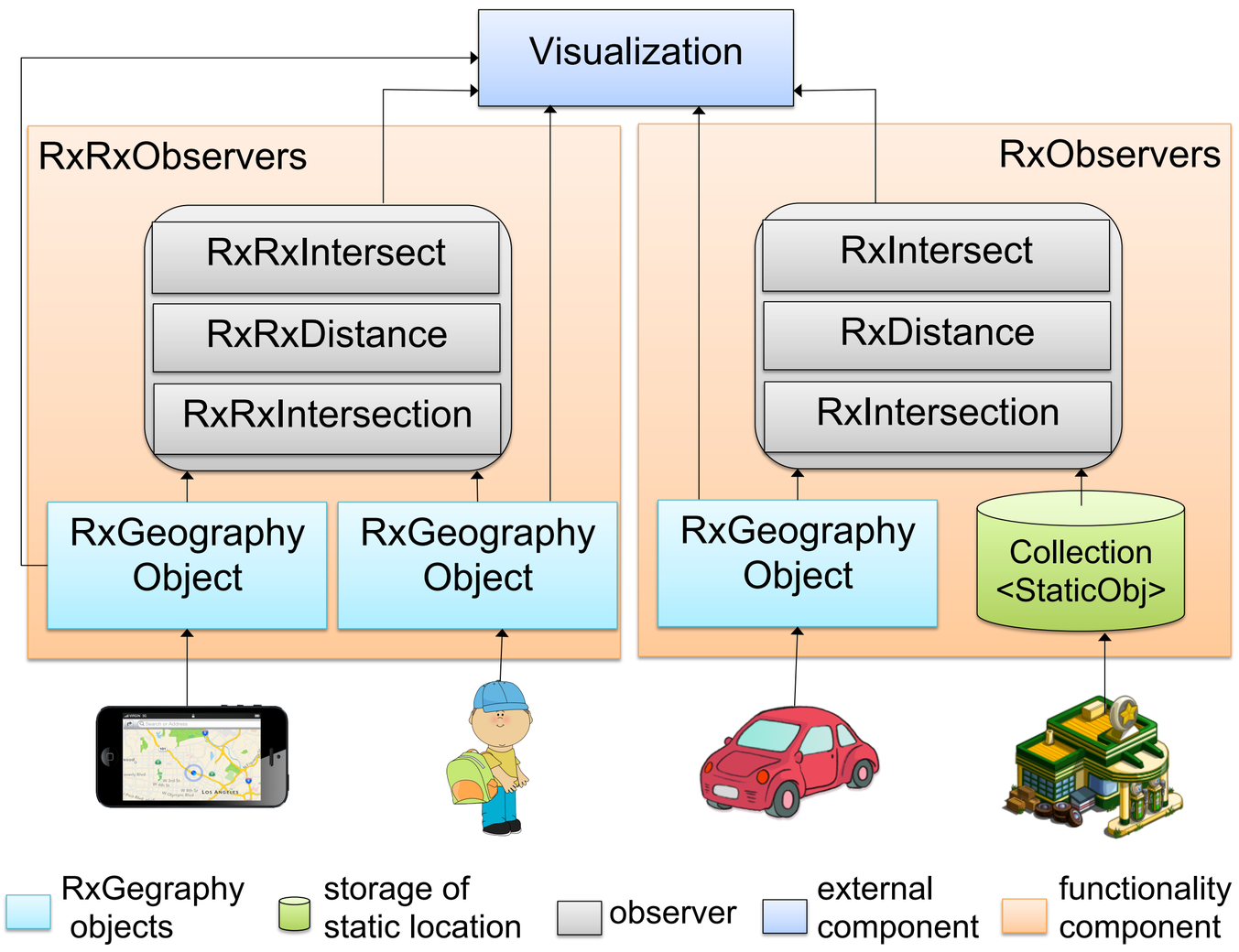
Vehicle Tracking for Insurance

Promises

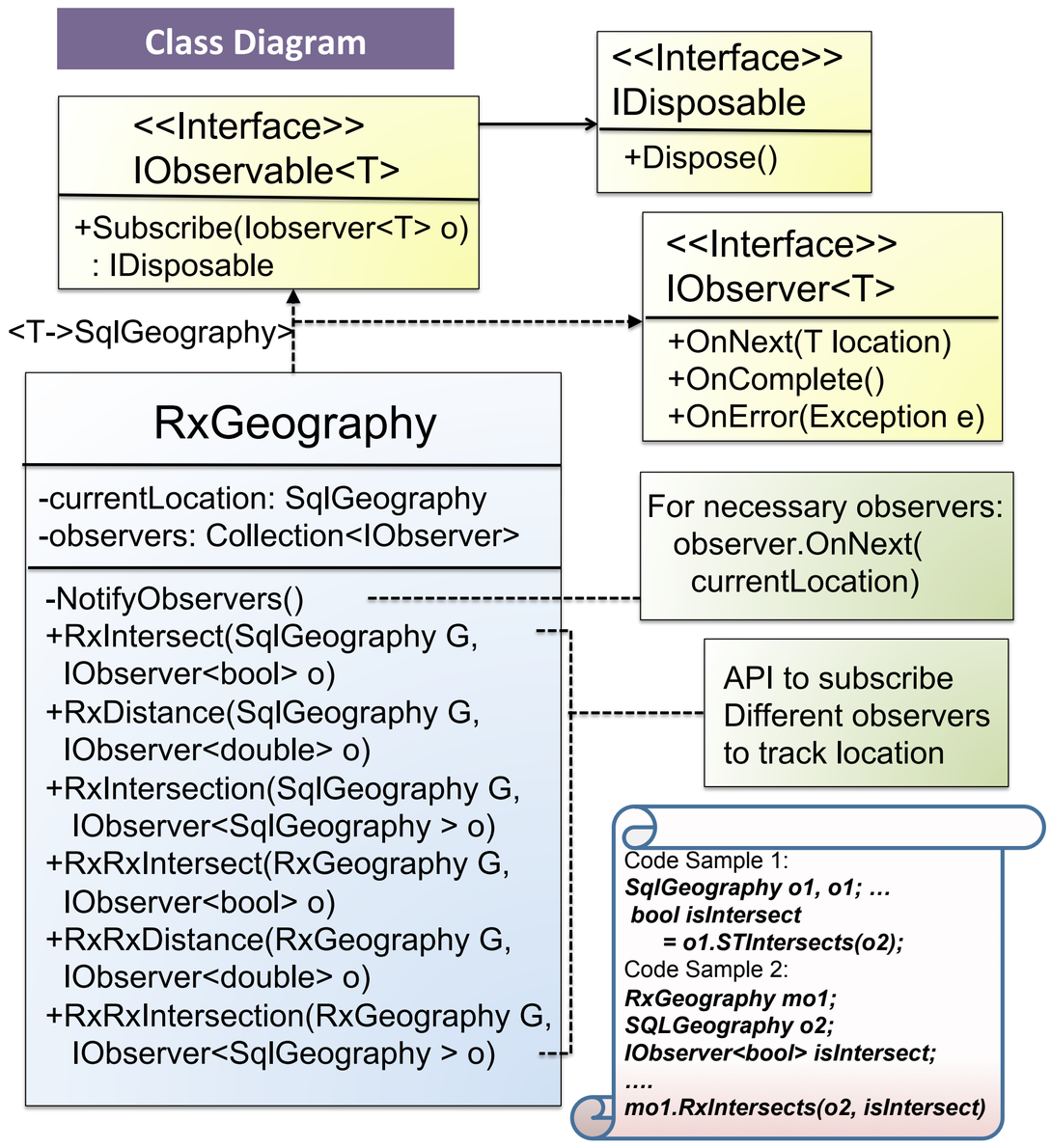


Technique

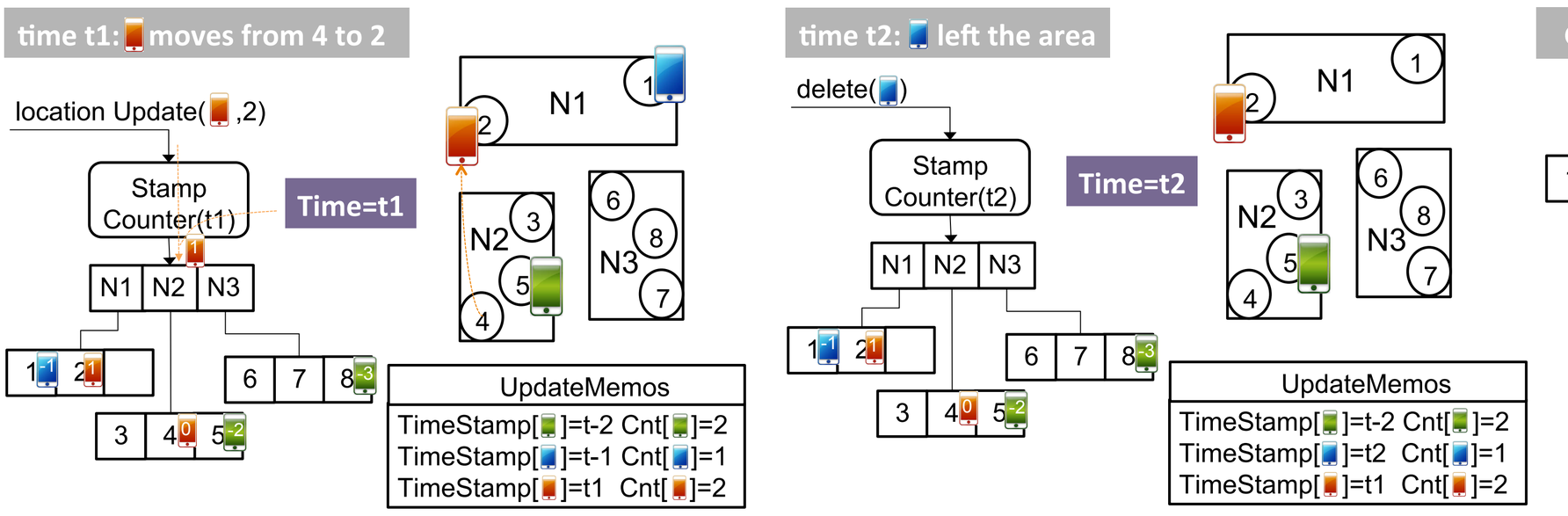
Architecture



Class Diagram

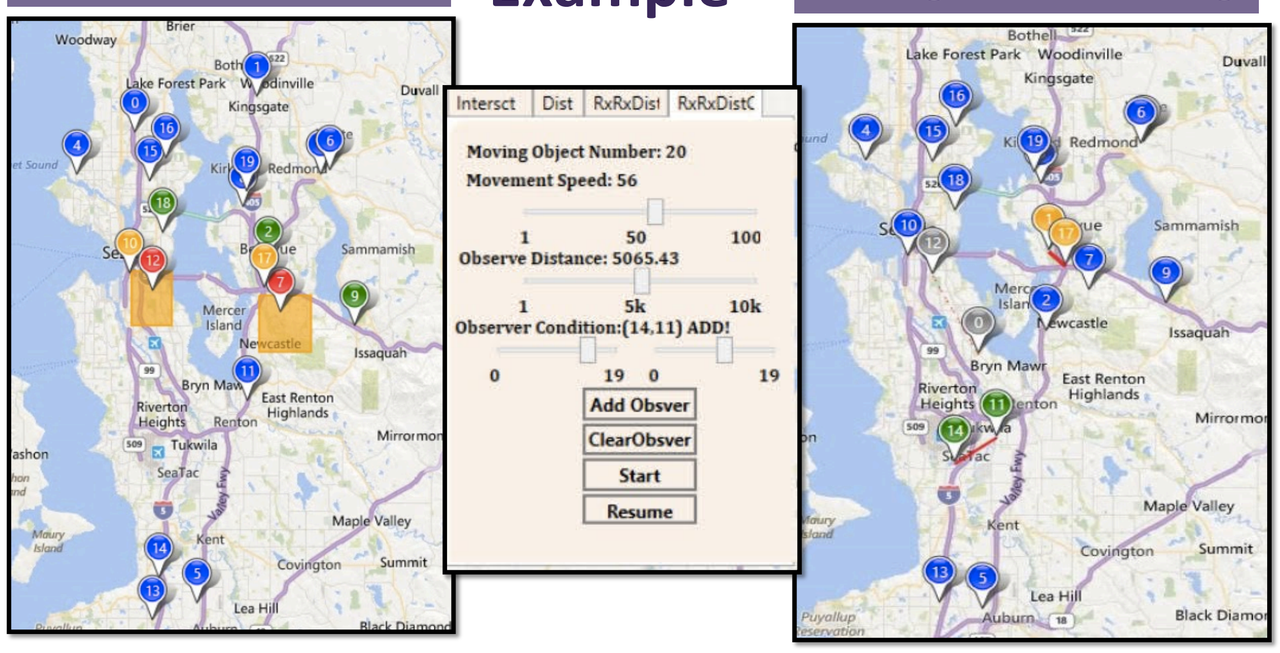


RUM-Tree



Example

Static observes Moving



Moving observes Moving

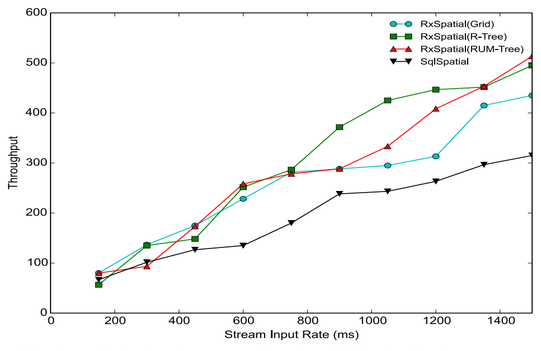
Criminal Activity Tracking
Intersect Status Tracking

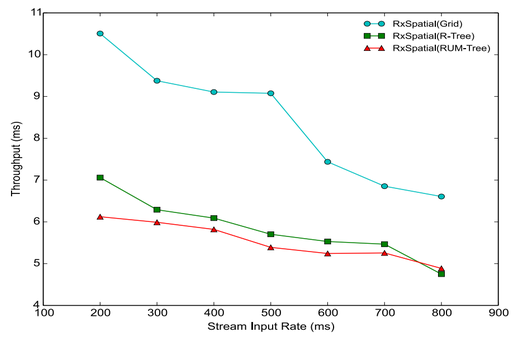
Control Panel

Collaborative Vehicle Proximity Tracking

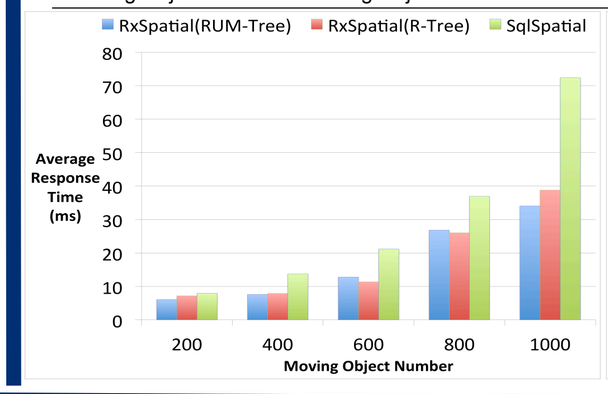
Performance Evaluation

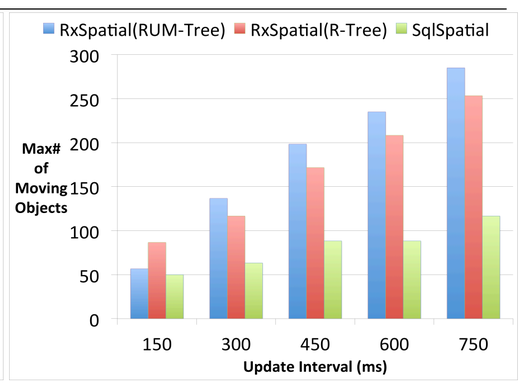
Static Object observing Moving object Performance





Moving Objects observer Moving Objects Performance





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